EART11200: Natural Scientist's Toolkit



This week will cover Primary Productivity
First we will look at the basic tools of fractions and factorising.

Fractions

- A way of expressing a portion of a whole e.g. third (⅓) an apple, eighth (⅛) of a cake
- Expressed as (numerator/denominator) where the numerator is the ‘whole’ divided by the denominator (number of parts by which the whole is divided) – in the above the whole cake is divided into eight equal slices.



Image from *The Guardian* puzzle page

Proper, improper and percentages

- Proper fractions are those where the numerator < denominator (e.g. 2/7)
- An improper fraction is one where the numerator > denominator (e.g. 7/3)
- An improper fraction can therefore also be written as a mixed fraction (2 ⅓), that is a whole number and a proper fraction
- One percent (%) is one hundredth (1/100) of a whole. 10% is 10/100 (or 1/10, one tenth) of the whole, and so on.

Combining fractions

- Multiplying $\frac{1}{4} x \frac{3}{7} x \frac{4}{9} = \frac{12}{252} = \frac{1}{21}$

Simplify by dividing both numerator and denominator by the same number (12 here)

- Dividing $\frac{1}{4} \div \frac{3}{7} = \frac{1}{4} x \frac{7}{3} = \frac{7}{12}$

Turn the divisor upside down and multiply

- Adding/Subtracting

$$\frac{1}{4} + \frac{5}{6} - \frac{11}{12} = \frac{3}{12} + \frac{10}{12} - \frac{11}{12} = \frac{2}{12} = \frac{1}{6}$$

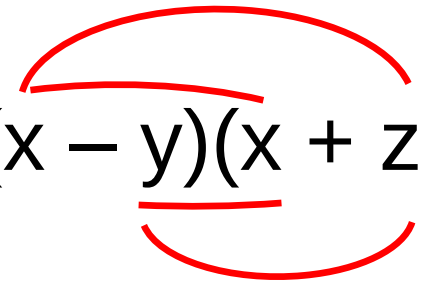
Write all fractions with a common denominator, then add/subtract the numerators. Simplify if required.

Expanding brackets

Remove the brackets:

$4(x+xy-3) = 4x + 4xy - 12$

$x(x^2 + xy + 2/x) = x^3 + x^2y + 2$



$(x - y)(x + z) = x^2 - yx + zx - yz$

$= x^2 + x(z-y) - yz$

Factorising

The reverse process of expanding brackets: look for common factor

$42y + 7 = 7(6y + 1)$ Common factor is 7

$6p^2 – 3p = 3p(2p – 1)$ Common factor 3p

$r^2 + 3r + 2 = (r + 2)(r + 1)$

$4v^2 + 12v – 40 = 4(v^2 + 3v – 10)$

$= 4(v + 5)(v – 2)$ Common factor 4, and terms in brackets

$ar^2 + br + c = (dr + e)(fr + g)$

$dxf=a$, and $exg=c$, and $dxg + exf = b$

Make sure you are comfortable with fractions and brackets/factorising, we will be using them in this week's problems.

If you find this difficult, ask one of the members of staff during the practical session.